

Finding Path from Dijkstra

- Maintain a parent array/vector along with distance array/vector
- Store parent information, when updating a neighbour node
- To get a path, traverse the parent array from destination towards source and store the vertices/nodes to a path vector
- Print the path vector from backwards

Solution approach for kth shortest path

- Change the type of distance array to array of a priority_queue /set.
- Update the distance of priority queue, with the new distance if the size of the priority queue is less than k
- If the priority queue size is k, then update the distance if the new distance is less than the largest element of the queue

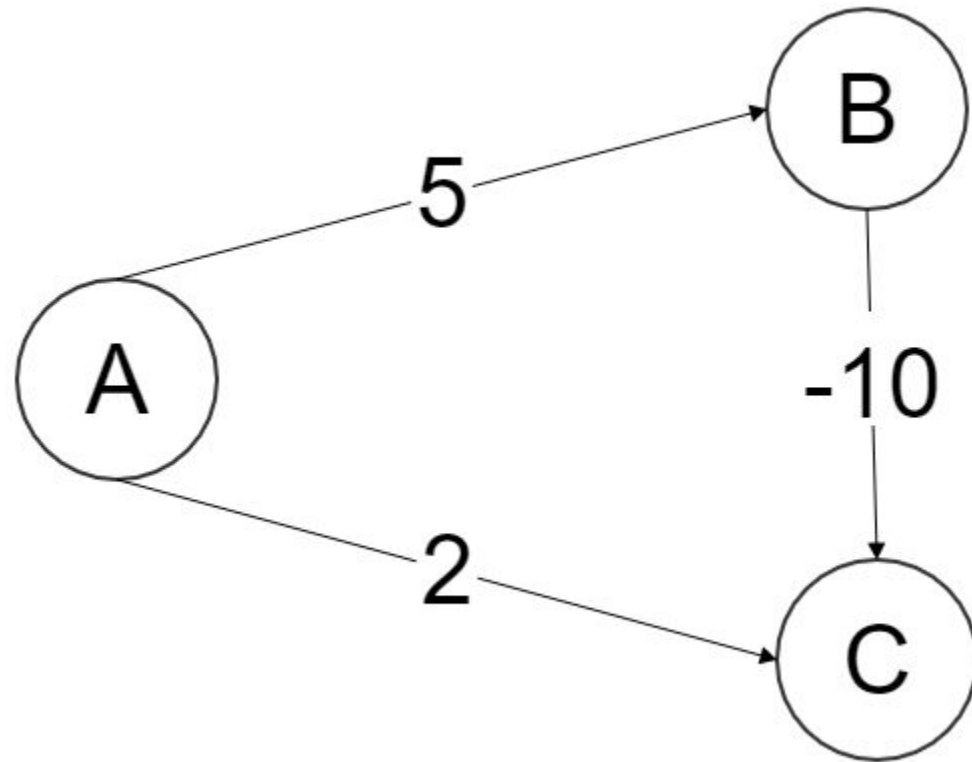
Problems involving kth shortest path

- CSES 1196
 - <https://cses.fi/problemset/task/1196>
- Lightoj 1099
 - <https://lightoj.com/problem/not-the-best>

Bellman Ford Algorithm

Can we apply Dijkstra to any weighted graph?

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Bellman Ford Algorithm

```
for v in V:
    v.distance = infinity
    v.p = None
source.distance = 0
for i from 1 to |V| - 1:
    for (u, v) in E:
        relax(u, v)
```

```
relax(u, v):
    if v.distance > u.distance + weight(u, v):
        v.distance = u.distance + weight(u, v)
        v.p = u
```

Time Complexity

- Initialize distance of all vertices $O(v)$
- Outer loop iteration: $v-1$
- Inner loop iteration: E

Total operation: $O((v-1)*E) = O(VE)$

Proof of Correctness

Goal:

- Prove $d[v] = \delta(s, v)$ after $|V| - 1$ iterations
- If there exists a negative-weight cycle reachable from s , then Bellman-Ford correctly detects it.

Proof of Correctness

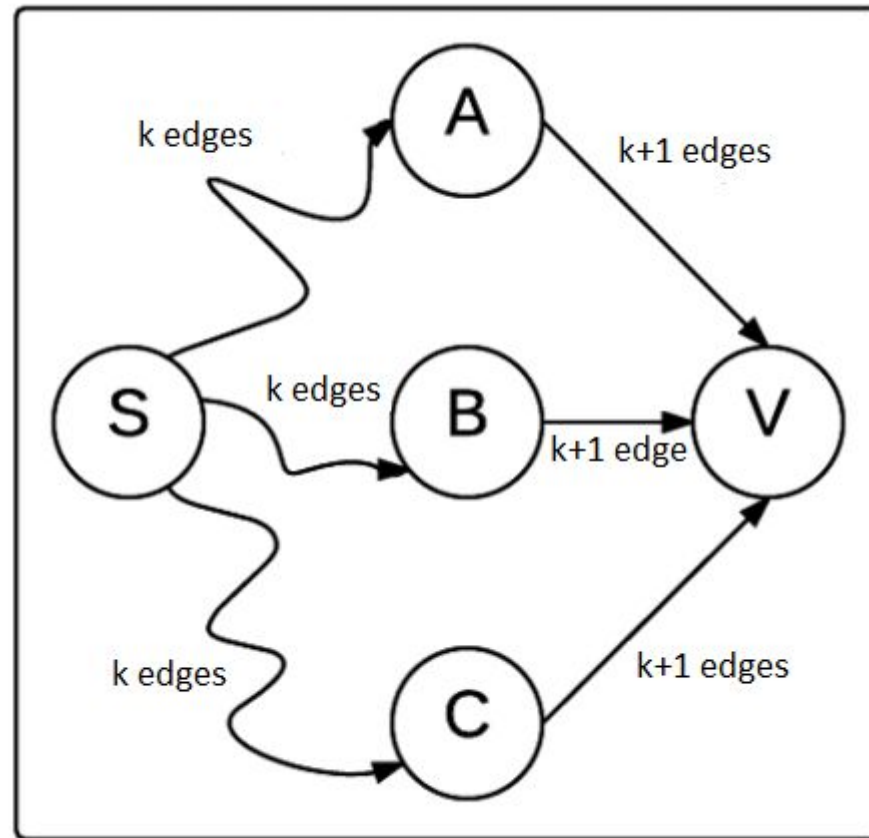
Base Case:

- The only vertex reachable with 0 edges is s , and
 $d[s] = 0 = \delta(s, s)$.

Inductive Hypothesis:

- Assume that after i iterations, for all vertices v reachable from s by a path of at most i edges:
 $d[v] = \delta(s, v)$.

Proof of Correctness



Proof of Correctness

Inductive step:

Let vertex v be reachable from s via a shortest path P with $i+1$ edges:

$$P=(s,\dots,u,v)$$

By inductive hypothesis, after i iterations:

$$d[u] = \delta(s,u)$$

Since $(u,v) \in E$, we relax it:

$$d[v] \leq d[u] + w(u,v)$$

$$d[v] \leq \delta(s,u) + w(u,v)$$

$$d[v] \leq \delta(s,v)$$

So, after $i+1$ iteration, $d[v]$ holds the correct shortest path with $i+1$ edges

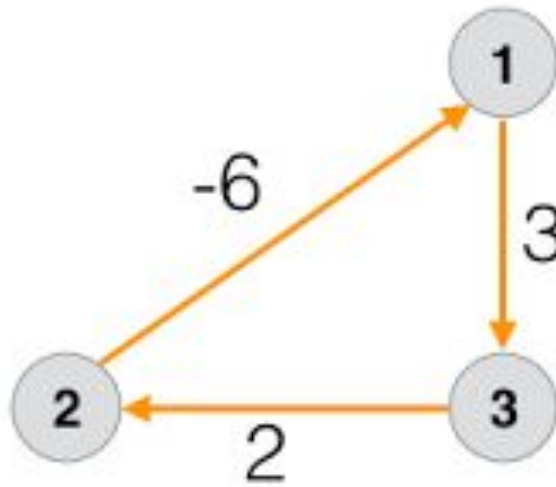
Proof of Correctness

Since the maximum number of edges in any shortest simple path is at most $|V|-1$, after $|V|-1$ iterations:

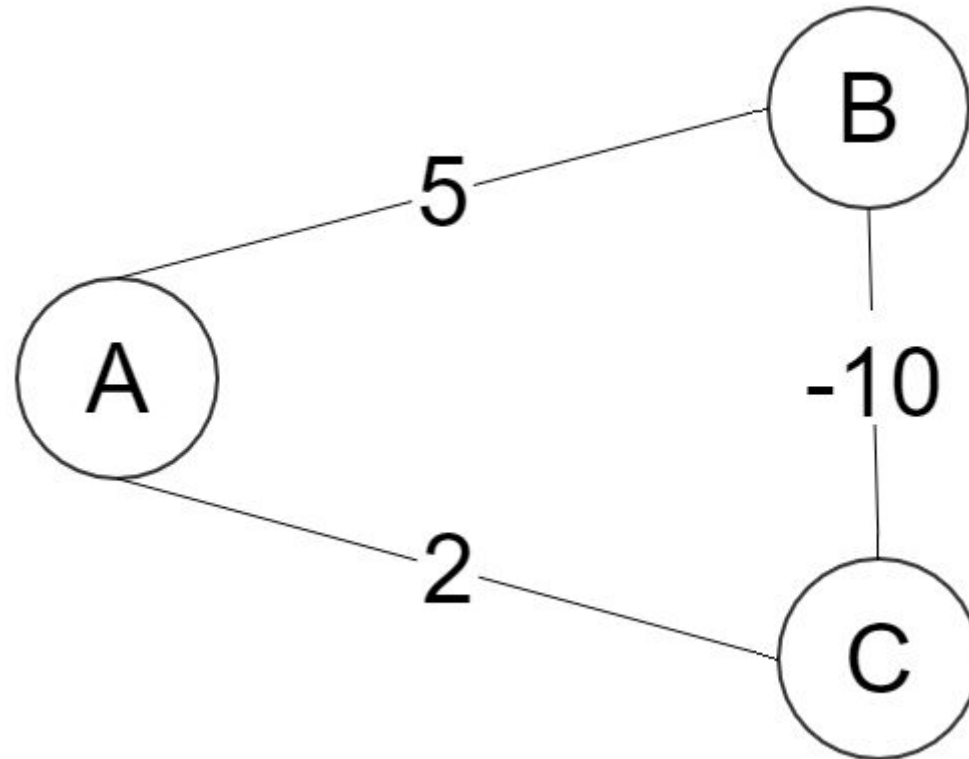
$d[v] = \delta(s, v)$ for all reachable vertices v .

Does bellman ford work on any graph?

Does bellman ford work on any graph?



Can we apply Dijkstra to any weighted graph?



Detecting Negative Cycle

```
for v in V:
    v.distance = infinity
    v.p = None
source.distance = 0
for i from 1 to |V| - 1:
    for (u, v) in E:
        relax(u, v)
for (u, v) in E:
    if v.distance > u.distance + weight(u, v):
        print "A negative weight cycle exists"
```

Practice Problem

- <https://basecamp.eolymp.com/en/problems/1453>
- <https://cses.fi/problemset/task/1673>
- <https://cses.fi/problemset/task/1197>